AUTOZ

Who are we?

A team that creates autonomous systems

Current Project

QUBI

360° Awareness in 3D

Qubi combines various sensors for localization, mapping and path planning for full environmental awareness and seamless navigation.

Move with Confidence

Equipped with a highly robust control system both at the higher and lower levels, Qubi is able to reach exactly where it needs, the way it needs to.

Designed for Versatility

Its CAD-engineered structure and size ensure smooth mobility across various terrains and spaces, from industrial sites to surveillance zones.

Reliable and Safe

With intelligent power management, real-time diagnostics, and emergency stop mechanisms, Qubi delivers secure and efficient performance.

Specifications

Autonomy Stack

SOFTWARE STACK
- ROS2 - Backbone for drivers and communication
- Custom EKF - Localization
- OctoMap - Probabilistic mapping
- OpenCV - Computer vision
- D* Lite - Dynamic Path Planner
- Pure Pursuit - Trajectory tracking

COMPUTING STACK
- Jetson Orin Nano - GPU Parallelization

SENSORS
- RPLIDAR S2 - 2D 360° Scanning
- Xsens IMU & GPS - Localization
- Encoders - Velocity estimation
- Intel RealSense D435i - Depth Perception

PERCEPTION AND CONTROL
- Xsens: IMU and GPS
- Depth Camera: For obstacle and lane detection

Electronics Stack

CONTROL SYSTEMS AND ACTUATORS
- Motors with inbuilt Encoders
- Cytron Motor Driver
- STM Bluepill - Implements PID for velocity control

POWER DISTRIBUTION
- Custom Power Distribution PCB designed in house
- Smaller Regulators: Provided for low voltage systems.
- Power Supply:A 24V LiPo powers 18V motors via a TDK regulator.

E-STOP AND DIAGNOSTICS
- Onboard Safety Systems: Equipped with emergency cut-off relays working in conjunction with sensors
- Diagnostics Board: Reports back key parameters to the onboard computer for monitoring
- Handheld E- Stop: Utilizes RF communication to ensure reliable operation

Mechanical Stack

HARDWARE
- Aluminium extrusion pipes 20mmx20mm
- L clamps-for connnecting extrusion pipes
- 10 inches wheels
- Custom Shafts and brackets
- Single Caster

ANALYSIS
- FEA-stimulate the load carrying capabilities

DESIGN
- CAD:fusion 360 and solid works-design chassis,shaft and brackets
- 3D printers-Additive Manufacturing

MANUFACTURING
- 3D printing- make mounts for electronic components and prototype

Past Projects

- Pratham; An autonomous ground robot
- Applications: Mine detection, delivery
- Predecessor to Project Qubi

- L3 autonomous vehicle
- Focus: Advanced automation for improved functionality, safety, and performance

- Fully autonomous hexacopter for surveillance
- Features: Equipped with cameras and control hardware

Our Team

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